# Restomate Backend API Documentation

## 📋 Daftar Isi



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## Overview

Restomate adalah sistem informasi berbasis LLM untuk manajemen restoran yang menyediakan analisis data, rekomendasi AI, dan insights bisnis komprehensif.

### Fitur Utama:

- ✅ **Menu Management** - CRUD operasi menu lengkap
- ✅ **AI-Powered Recommendations** - Rekomendasi paket makanan dengan GROQ API
- ✅ **Analytics Dashboard** - Analisis penjualan dan performa
- ✅ **Customer Feedback** - Manajemen feedback dengan sentiment analysis
- ✅ **Stock Predictions** - Prediksi stok berbasis AI
- ✅ **Promotion Generation** - Rekomendasi promosi otomatis
- ✅ **Menu Trends** - Analisis tren menu

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## Setup & Konfigurasi

### Environment Variables (.env)

```env
```

```
# Database Configuration
MYSQL_HOST=localhost
MYSQL_PORT=8889
MYSQL_USER=root
MYSQL_PASSWORD=root
MYSQL_DATABASE=restomate

# GROQ API Configuration
GROQ_API_KEY=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

# Next.js Configuration
NEXTAUTH_SECRET=your_nextauth_secret_here
NEXTAUTH_URL=http://localhost:3000
```

### Dependencies

json

```
{
  "mysql2": "^3.6.0",
  "next": "^14.0.0",
  "clsx": "^2.0.0",
  "tailwind-merge": "^2.0.0"
}
```

### GROQ Model Update

⚠️ **PENTING**: Model `llama-3.1-70b-versatile` sudah didecommissioned. Semua API sekarang menggunakan `llama3-8b-8192`.

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## Struktur Database

### Tabel Utama:

sql

```
-- Restoran
RESTAURANT (id_restaurant, email, password)

-- Menu Items
menu (Id_Menu, Nama_Menu, Harga, Kategori, Deskripsi, id_restaurant)

-- Customer Orders
Customer (Invoice_Id, Tanggal_Order, Harga_Total, id_restaurant)

-- Order Items
MEMESAN_MENU (id_menu, kuantitas, id_customer)

-- Food Packages
PAKET (id_paket, id_menu, id_restaurant)

-- Stock Management
STOK (id_stok, nama_bahan, kuantitas, tanggal_pembelian, tanggal_exp, id_menu, id_restaurant)

-- Customer Feedback
CUSTOMER_FEEDBACK (id_feedback, id_customer, id_restaurant, rating, feedback_text, feedback_date)
```

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## API Endpoints

### 🍽️ Menu Management (`/api/menu`)

#### GET - Fetch Menu Items & Food Packs

```http
GET /api/menu?restaurant_id=1
```

**Response:**

```json
```

```
{
  "success": true,
  "data": {
    "menuItems": [...],
    "foodPacks": [...],
    "summary": {
      "totalMenuItems": 25,
      "totalFoodPacks": 10,
      "categories": ["Makanan Utama", "Minuman", "Snack"],
      "priceRange": { "min": 5000, "max": 50000 }
    }
  }
}
```

### POST - Add New Menu Item

```http
POST /api/menu
Content-Type: application/json

{
  "name": "Nasi Gudeg Special",
  "price": 25000,
  "category": "Makanan Utama",
  "description": "Gudeg khas Yogyakarta dengan ayam",
  "restaurant_id": 1
}
```

### PUT - Update Menu Item

```http
PUT /api/menu
Content-Type: application/json

{
  "id": 1,
  "name": "Nasi Gudeg Premium",
  "price": 30000,
  "category": "Makanan Utama",
  "description": "Updated description"
}
```

### DELETE - Remove Menu Item

```http
DELETE /api/menu?id=1
```

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## 🤖 AI Food Pack Recommendations (`/api/menu/food-packs`)

### GET - Generate AI Recommendations

```http
GET /api/menu/food-packs?restaurant_id=1&type=recommendations
```

**Response:**


```json
{
  "success": true,
  "data": {
    "packs": [
      {
        "id": "ai_pack_1234567890_0",
        "name": "Paket Hemat Spesial",
        "description": "Kombinasi strategis menu terpilih",
        "items": ["Nasi Gudeg", "Es Teh Manis"],
        "price": 27000,
        "originalPrice": 30000,
        "discountPercent": 10,
        "reasoning": "Paket ini menggabungkan makanan utama populer",
        "estimatedDemand": "High",
        "category": "Value Pack",
        "generated": true
      }
    ]
  }
}
```


**Features:**

- 🔄 **Retry Mechanism**: 3 attempts sebelum fallback
- 🧠 **LLM Analysis**: Menggunakan data penjualan real
- 📊 **Data-Driven**: Basis rekomendasi dari performa menu
- 🛡️ **Fallback System**: Enhanced algorithmic generation

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## 📊 Analytics Dashboard (`/api/analytics`)

### GET - Comprehensive Analytics

```http
GET /api/analytics?restaurant_id=1&period=30&start_date=2025-01-01&end_date=2025-01-31
```

### Query Parameters:

- `restaurant_id`: ID restoran (default: 1)
- `period`: Periode hari (default: 30)
- `start_date`: Tanggal mulai (optional)
- `end_date`: Tanggal akhir (optional)

### Response Data:

```json
{
  "success": true,
  "data": {
    "totalRevenue": 15750000,
    "totalOrders": 847,
    "averageOrderValue": 18599,
    "topSellingItems": [...],
    "revenueByCategory": [...],
    "dailySales": [...],
    "monthlySales": [...],
    "customerInsights": {
      "uniqueCustomers": 523,
      "avgOrderValue": 18599,
      "ordersPerCustomer": 1.62
    },
    "inventoryStatus": [...]
  }
}
```

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## 💬 Customer Feedback (`/api/feedback`)

### GET - Fetch Feedback with Analysis

```http
GET /api/feedback?restaurant_id=1&page=1&limit=20&rating=5&period=30
```

**Features:**

- 🎭 **Sentiment Analysis**: Otomatis menganalisis sentimen feedback
- 📈 **Trend Analysis**: Tracking rating bulanan
- 🔍 **Advanced Filtering**: Filter by rating, status, periode
- 📊 **Summary Statistics**: Rating distribution dan insights

### POST - Add New Feedback

```http
POST /api/feedback
Content-Type: application/json

{
  "customer_id": 123,
  "restaurant_id": 1,
  "rating": 5,
  "feedback_text": "Makanan sangat enak dan pelayanan ramah!",
  "status": "pending"
}
```

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## 📦 Orders Management (`/api/orders`)

### GET - Fetch Orders

```http
GET /api/orders?restaurant_id=1&page=1&limit=20&include_items=true&period=30
```

**Response Features:**

- 📋 **Order Details**: Lengkap dengan item breakdown
- 📊 **Summary Stats**: Total orders, revenue, popular items
- 📈 **Today's Performance**: Orders dan revenue hari ini
- 🕐 **Recent Orders**: 5 order terakhir

### POST - Create New Order

```http
```

```
POST /api/orders
Content-Type: application/json

{
  "restaurant_id": 1,
  "items": [
    { "menu_id": 1, "quantity": 2 },
    { "menu_id": 3, "quantity": 1 }
  ],
  "total_amount": 75000,
  "customer_info": {...}
}
```

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## 🎯 AI Promotions (`/api/promotions`)

### GET - Generate Promotion Recommendations

```
http

GET /api/promotions?restaurant_id=1&type=generate
```

### AI Analysis Meliputi:

- 📊 Menu performance data
- 🎯 Customer feedback analysis
- 💰 Revenue optimization strategies
- 🔄 Market trends consideration

### Response:

```
json
```

```
{
  "success": true,
  "data": {
    "promotions": [
      {
        "type": "Bundle Deal",
        "description": "Paket hemat menu populer + menu kurang laris",
        "reasoning": "Meningkatkan penjualan menu underperforming",
        "estimatedImpact": "+25% sales",
        "details": "Diskon 20% untuk pembelian paket"
      }
    ]
  }
}
```

### POST - Apply Promotion

```
http

POST /api/promotions
Content-Type: application/json

{
  "promotion": {
    "type": "Happy Hour",
    "description": "Diskon 30% minuman jam 14-16",
    "reasoning": "Increase traffic during slow hours"
  },
  "restaurantId": 1
}
```

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## 📈 Menu Trends Analysis (`/api/menu-trends`)

### GET - AI-Powered Trend Analysis

```
http

GET /api/menu-trends?restaurant_id=1
```

### Response:

```
json
```

```
{
  "success": true,
  "data": {
    "trends": [
      {
        "trend": "rising",
        "itemName": "Nasi Gudeg",
        "currentSales": 150,
        "predictedSales": 180,
        "growthRate": 20,
        "confidence": 85,
        "reasoning": "Menu dengan performa terbaik",
        "recommendations": [
          "Pertahankan kualitas dan konsistensi",
          "Pertimbangkan variasi atau bundle deals"
        ],
        "category": "Makanan Utama",
        "seasonality": "Stabil sepanjang tahun"
      }
    ]
  }
}
```


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### 📦 Stock Predictions (`/api/generate-stock-predictions`)

#### GET - AI Stock Predictions

```
http

GET /api/generate-stock-predictions?restaurant_id=1&period=week
```

**Parameters:**

- `period`: `week` | `month` | `quarter`

**Features:**

- 🤖 **Pure LLM Analysis**: Menggunakan GROQ AI
- 📊 **Comprehensive Data**: All-time stock dan sales data
- ⏰ **Period-Aware**: Analisis sesuai periode yang dipilih
- 🎯 **Actionable Insights**: Rekomendasi konkret

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## LLM Integration

## GROQ API Configuration

typescript

```typescript
// lib/llm.ts
const GROQ_API_URL = 'https://api.groq.com/openai/v1/chat/completions';

const response = await fetch(GROQ_API_URL, {
  method: 'POST',
  headers: {
    'Authorization': `Bearer ${process.env.GROQ_API_KEY}`,
    'Content-Type': 'application/json',
  },
  body: JSON.stringify({
    model: 'llama3-8b-8192', // Updated model
    messages: [...],
    temperature: 0.7,
    max_tokens: 1500
  })
});
```

## Fallback System

typescript

```typescript
// Enhanced Fallback Strategy
try {
  // Try LLM first (3 attempts)
  const llmResult = await generateLLMRecommendations();
  return llmResult;
} catch (error) {
  // Fallback to algorithmic analysis
  return generateAlgorithmicFallback();
}
```

## Rate Limiting

typescript

```
class SimpleRateLimiter {
  private lastRequest = 0;
  private minDelay = 3000; // 3 seconds

  canMakeRequest(): boolean {
    const now = Date.now();
    return (now - this.lastRequest) >= this.minDelay;
  }
}
```

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## Error Handling

### Standard Error Response

```
json

{
  "success": false,
  "error": "Error type",
  "message": "Detailed error message",
  "data": null
}
```

### HTTP Status Codes

- `200` - Success
- `400` - Bad Request (validation errors)
- `404` - Not Found (no data)
- `500` - Internal Server Error

### Logging System

```
typescript

// Comprehensive logging
console.log('🍽️ Fetching menu data...');
console.log('✅ Success: Retrieved 25 items');
console.error('❌ Error:', error.message);
console.warn('⚠️ Warning: Fallback used');
```

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## Testing

### Environment Setup

```bash
# 1. Clone repository
git clone <repository-url>

# 2. Install dependencies
npm install

# 3. Setup database
mysql -u root -p restomate < scripts/restomate.sql

# 4. Configure environment
cp .env.example .env
# Edit .env with your configurations

# 5. Run development server
npm run dev
```

## API Testing

```bash
# Test menu endpoint
curl -X GET "http://localhost:3000/api/menu?restaurant_id=1"

# Test AI recommendations
curl -X GET "http://localhost:3000/api/menu/food-packs?restaurant_id=1&type=recommendations"

# Test analytics
curl -X GET "http://localhost:3000/api/analytics?restaurant_id=1&period=30"
```

## Database Verification

```sql
```

```sql
-- Check data availability
SELECT COUNT(*) FROM Customer WHERE id_restaurant = 1;
SELECT COUNT(*) FROM menu WHERE id_restaurant = 1;
SELECT COUNT(*) FROM STOK WHERE id_restaurant = 1;

-- Verify relationships
SELECT c.Invoice_Id, c.Harga_Total, m.Nama_Menu
FROM Customer c
JOIN MEMESAN_MENU mm ON c.Invoice_Id = mm.id_customer
JOIN menu m ON mm.id_menu = m.Id_Menu
WHERE c.id_restaurant = 1
LIMIT 5;
```

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## Production Checklist

### Security

- [ ] API key protection
- [ ] Input validation
- [ ] SQL injection prevention
- [ ] Rate limiting implementation

### Performance

- [ ] Database indexing
- [ ] Query optimization
- [ ] Caching strategy
- [ ] Connection pooling

### Monitoring

- [ ] Error tracking
- [ ] Performance metrics
- [ ] API usage analytics
- [ ] Health checks

### Deployment

- [ ] Environment variables secured
- [ ] Database migrations
- [ ] SSL certificates
- [ ] Load balancing

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## Support & Maintenance

### Regular Tasks

1. **Monthly**: Review API usage dan performance
2. **Weekly**: Check error logs dan fallback frequency
3. **Daily**: Monitor database performance
4. **Real-time**: Alert system untuk critical errors

## Troubleshooting Common Issues

### GROQ API Errors

```bash
# Check API key
echo $GROQ_API_KEY

# Test connectivity
curl -H "Authorization: Bearer $GROQ_API_KEY" \
  https://api.groq.com/openai/v1/models
```

### Database Connection Issues

```bash
# Test connection
mysql -h localhost -P 8889 -u root -p restomate -e "SELECT 1"

# Check running processes
ps aux | grep mysql
```

### Performance Issues

```sql
-- Check slow queries
SHOW PROCESSLIST;

-- Analyze table performance
EXPLAIN SELECT * FROM Customer WHERE id_restaurant = 1;
```

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